Advanced Machine Learning, ETH Zurich

Hinge-chain networks

Any piecewise-linear function, one kink per layer

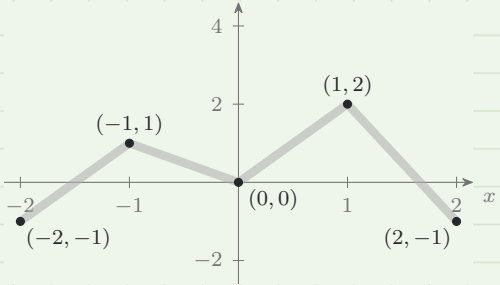
Carlos Cotrini

Autumn semester 2026

Entry 1

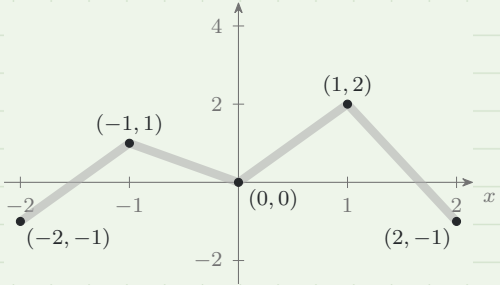
The goal

Our target is the zigzag f, which joins five points with straight lines on the interval from -2 to 2. We want a neural network that takes a number x and outputs $y = f(x)$, for every x in the interval.



Grey: the target f, the zigzag.

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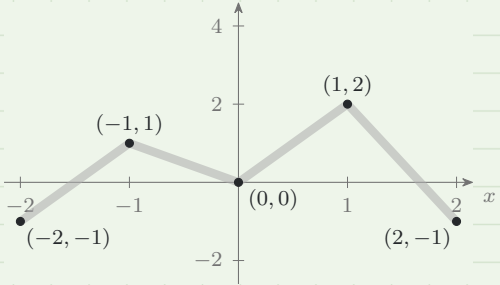
Grey: the target f, the zigzag.



In goes x, out comes y. We ask for $y = f(x)$ at every x from -2 to 2.

Our goal: a network whose output is exactly the zigzag

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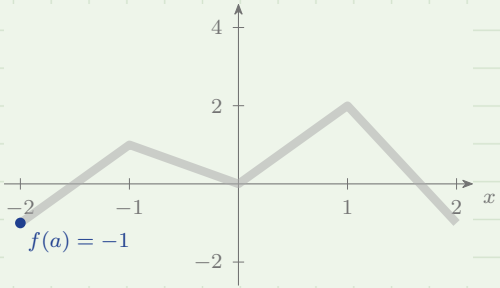
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In goes x, out comes y. We ask for $y = f(x)$ at every x from -2 to 2.

Exactly: an error of zero everywhere. We construct the network from ReLUs, piece by piece, with no training.

Start at the left end $a = -2$, at the height $f(a) = -1$, and walk to the right along a straight line. At a *kink* the slope changes, and the walk goes on straight again. The zigzag has three kinks, t_1, t_2 and t_3.



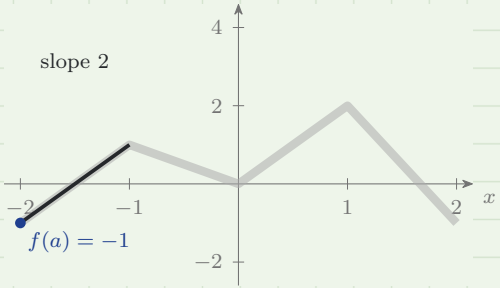
Grey: the zigzag. Graphite: the walk, one piece per click.

start value $f(a) = -1$

slopes

kinks

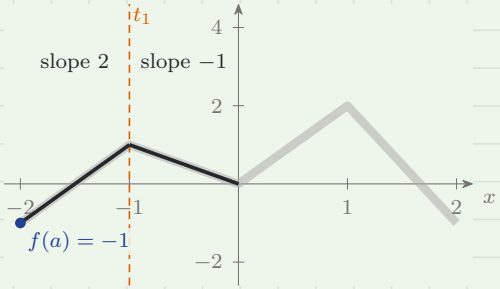
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Grey: the zigzag. Graphite: the walk, one piece per click.

start value	$f(a) = -1$
slopes	2
kinks	

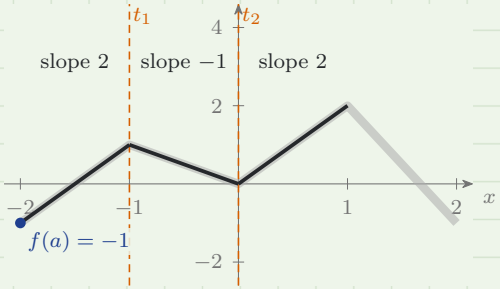
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Grey: the zigzag. Graphite: the walk, one piece per click.

start value	$f(a) = -1$
slopes	$2, -1$
kinks	$t_1 = -1$

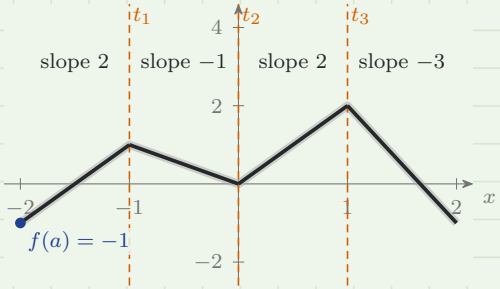
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start value	$f(a) = -1$
slopes	$2, -1, 2$
kinks	$t_1 = -1, t_2 = 0$

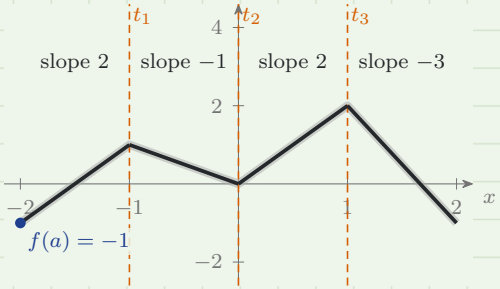
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slopes	$2, -1, 2, -3$
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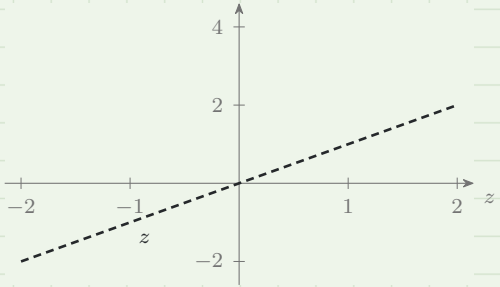
start value	$f(a) = -1$
slopes	$2,\ -1,\ 2,\ -3$
kinks	$t_1 = -1,\ t_2 = 0,\ t_3 = 1$

Such a function is *piecewise linear*: straight pieces that meet at kinks. Its start value, its slopes and its kinks pin it down. We build a network from exactly these numbers.

Entry 2

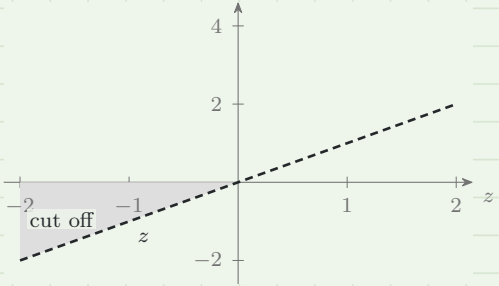
The hinge

The rectified linear unit, ReLU, is the activation of every neuron in our network. It passes a positive number on as it is and turns a negative number into 0: whatever falls below the floor at zero is cut off.



Dashed: the number z that comes in, plotted against itself.

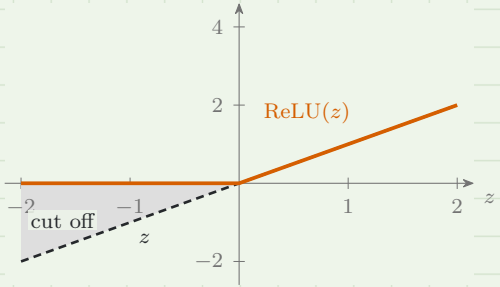
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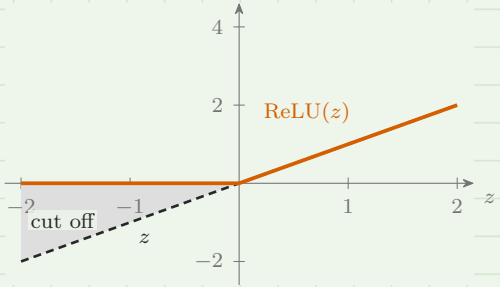


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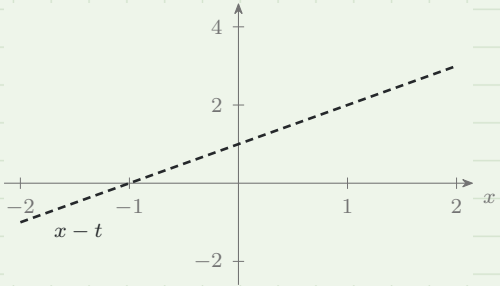
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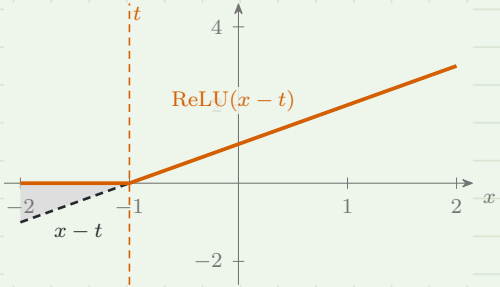
$$\mathrm{ReLU}(z) = \max\{z, 0\}$$

Feed the ReLU the input x minus a number t. Left of t, the number $x-t$ is negative and the floor sets it to 0; right of t, it rises with slope 1. We call $\mathrm{ReLU}(x-t)$ a *hinge*, and t its kink.



Dashed: the input shifted by t, the line $x-t$, here with $t=-1$.

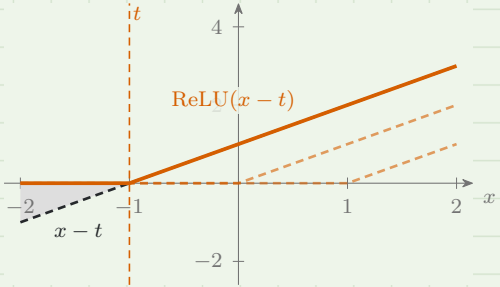
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The floor cuts what lies left of t. What is left is a hinge: flat at 0 up to its kink t, then rising with slope 1.

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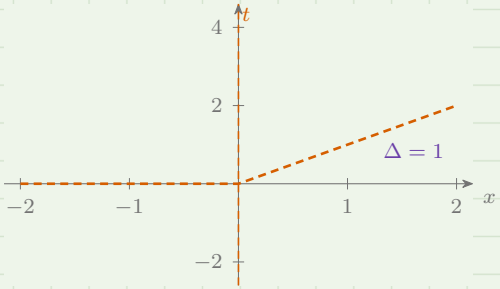


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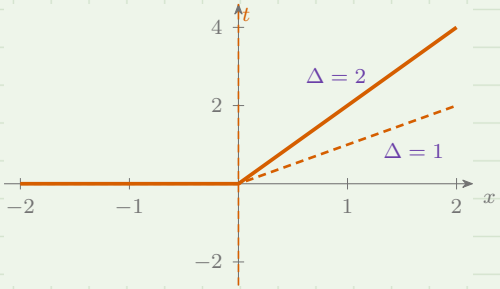
Changing t slides the hinge along x. Dashed: the same hinge at $t=0$ and at $t=1$.

Multiply a hinge by a number Δ. Up to its kink it stays 0; after the kink its slope is Δ in place of 1. A positive Δ bends it up, a negative one bends it down.



Dashed: the plain hinge, $\Delta = 1$.

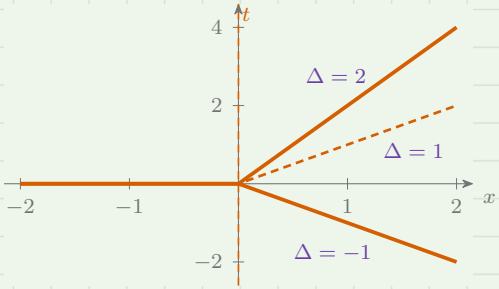
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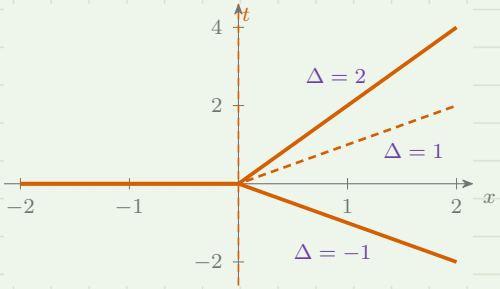


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$\Delta = 2$: after the kink, the slope is 2.

$\Delta = -1$: after the kink, the slope is -1, and the hinge bends down.

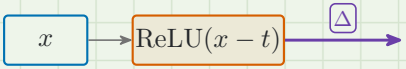
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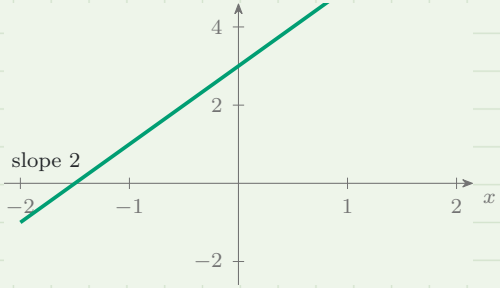
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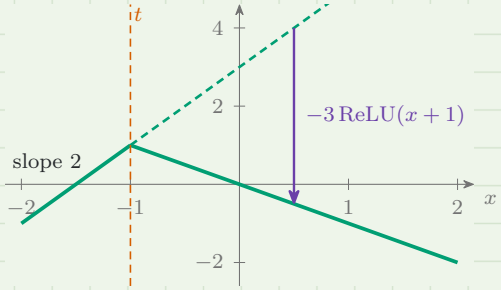
In a network, Δ is the weight on the arrow that leaves the neuron, so it acts after the ReLU.

Add $\Delta\,\mathrm{ReLU}(x-t)$ to a line. Left of t the hinge is 0, so nothing changes; right of t the slope grows by Δ. Here a line of slope 2 gets the hinge $-3\,\mathrm{ReLU}(x+1)$.



A line of slope 2.

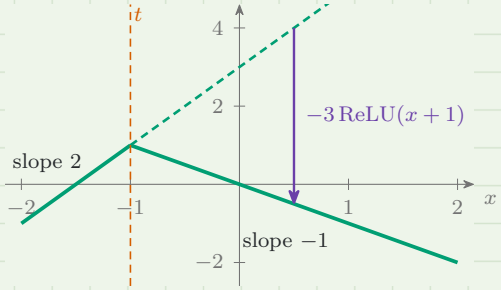
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A line of slope 2.

Add the hinge $-3\,\mathrm{ReLU}(x+1)$: its kink is $t=-1$ and its weight $\Delta=-3$. Left of -1 it adds nothing; right of -1 it pulls the line down (violet arrow).

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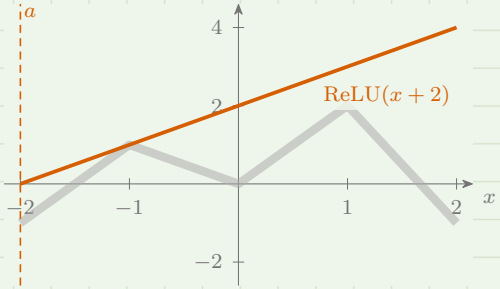
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Right of the kink the slope is $2+(-3)=-1$. The line now bends at t, and Δ is the change of slope there: a weighted hinge is a bend.

Entry 3

The hinge form

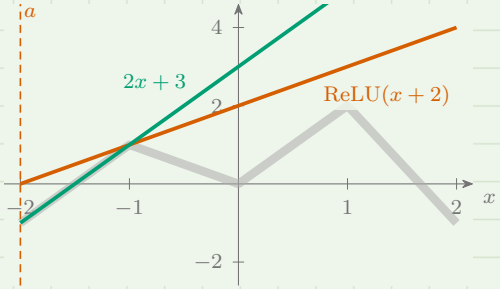
Put the kink of a hinge at the left end a. On $[a, b]$, with $b = 2$, the number $x - a$ is never negative, so the hinge $\mathrm{ReLU}(x - a) = x - a$ is a straight line. Times the first slope $\Delta_0 = 2$, plus $f(a) = -1$, it draws the first piece.



Grey: the zigzag. Vermillion: hinge 0.

Hinge 0 has its kink at the left end, $a = -2$. Right of a it never meets the floor: on $[a, b]$ it is the line $x + 2$, with slope 1.

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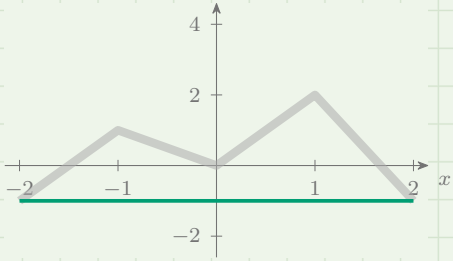


Grey: the zigzag. Vermillion: hinge 0. Green: the start value plus hinge 0.

Hinge 0 has its kink at the left end, $a = -2$. Right of a it never meets the floor: on $[a, b]$ it is the line $x + 2$, with slope 1.

Times the first slope $\Delta_0 = 2$, plus the start value -1: the line $2x + 3$. It is the first piece of the zigzag, continued straight on.

Start from $f(a) = -1$ and add hinge 0 for the first slope. Then add one hinge at each kink, weighted by the slope change there: the new slope minus the old one.

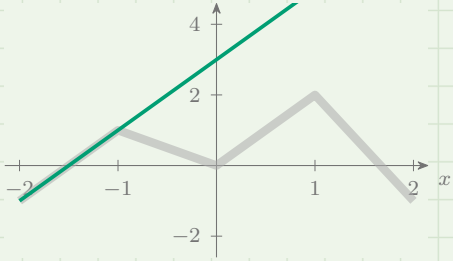


Grey: the zigzag. Green: the sum so far, one hinge more per click.

$$f(x) = -1$$

start value $f(a)$

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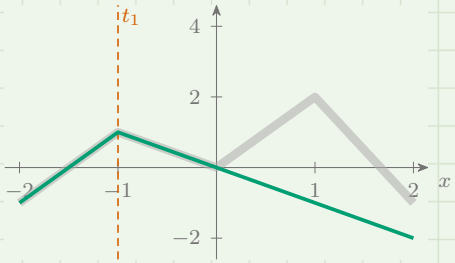


Grey: the zigzag. Green: the sum so far, one hinge more per click.

$$f(x) = -1 \qquad \text{start value } f(a)$$
$$+2\ \mathrm{ReLU}(x+2) \qquad \text{hinge 0: slope 0 to 2}$$

The zigzag is its start value plus one hinge per kink

Start from $f(a) = -1$ and add hinge 0 for the first slope. Then add one hinge at each kink, weighted by the slope change there: the new slope minus the old one.



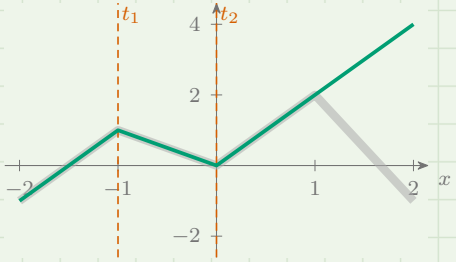
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$$f(x) = -1 \qquad \text{start value } f(a)$$

$$+2\ \mathrm{ReLU}(x+2) \qquad \text{hinge 0: slope 0 to 2}$$

$$-3\ \mathrm{ReLU}(x+1) \qquad \text{kink } -1\text{: slope 2 to } -1$$

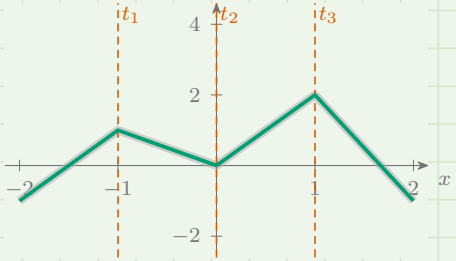
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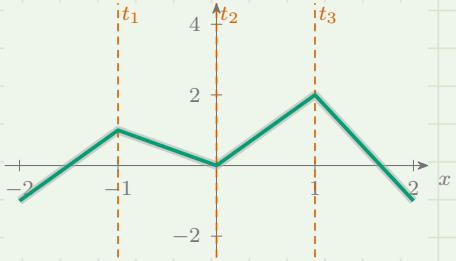
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$$
\begin{aligned}
f(x) = {} & -1 && \text{start value } f(a) \\
& +2\,\mathrm{ReLU}(x+2) && \text{hinge 0: slope 0 to 2} \\
& -3\,\mathrm{ReLU}(x+1) && \text{kink } -1\text{: slope 2 to } -1 \\
& +3\,\mathrm{ReLU}(x) && \text{kink 0: slope } -1 \text{ to } 2 \\
& -5\,\mathrm{ReLU}(x-1) && \text{kink 1: slope 2 to } -3
\end{aligned}
$$

Start from $f(a) = -1$ and add hinge 0 for the first slope. Then add one hinge at each kink, weighted by the slope change there: the new slope minus the old one.



Grey: the zigzag. Green: the sum so far, one hinge more per click.

$$
\begin{aligned}
f(x) = {} & -1 && \text{start value } f(a) \\
& +2\ \mathrm{ReLU}(x+2) && \text{hinge 0: slope 0 to 2} \\
& -3\ \mathrm{ReLU}(x+1) && \text{kink } -1\text{: slope 2 to } -1 \\
& +3\ \mathrm{ReLU}(x) && \text{kink 0: slope } -1 \text{ to } 2 \\
& -5\ \mathrm{ReLU}(x-1) && \text{kink 1: slope 2 to } -3
\end{aligned}
$$

The violet numbers are the slope changes, $\Delta = (2, -3, 3, -5)$: the first slope, then the change at each kink.

Take any continuous piecewise-linear f on $[a, b]$ with k kinks $t_1 < \cdots < t_k$. It is its start value plus $k + 1$ hinges: hinge 0 at $t_0 = a$, which gives the first slope, and one hinge at each kink, which bends the curve there.

$$f(x) = \underbrace{f(a)}_{\text{start value}} + \underbrace{\Delta_0 \,\mathrm{ReLU}(x - t_0)}_{\text{hinge 0}} + \underbrace{\Delta_1 \,\mathrm{ReLU}(x - t_1) + \cdots + \Delta_k \,\mathrm{ReLU}(x - t_k)}_{\text{one hinge per kink}}$$

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Δ_0 is the first slope, and Δ_ℓ is the slope change at the kink t_ℓ: the new slope minus the old one.

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The zigzag: $k = 3$ kinks at -1, 0, 1; $f(a) = -1$; $\Delta = (2, -3, 3, -5)$.

Why it holds: both sides start at $f(a)$, and on the piece after kink t_ℓ both have the slope $\Delta_0 + \Delta_1 + \cdots + \Delta_\ell$.

Entry 4

One kink per layer

A layer with a single ReLU neuron turns the number u it receives into $\mathrm{ReLU}(wu + c)$. That map only rises (when $w \geq 0$) or only falls (when $w \leq 0$), and a chain of such maps still only rises or only falls.



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The zigzag rises, falls, rises and falls again: no chain of single neurons can draw it. We give each layer two neurons.

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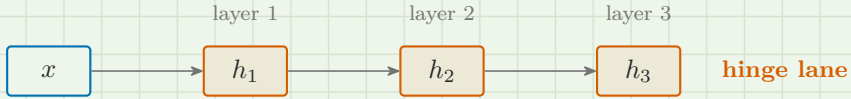


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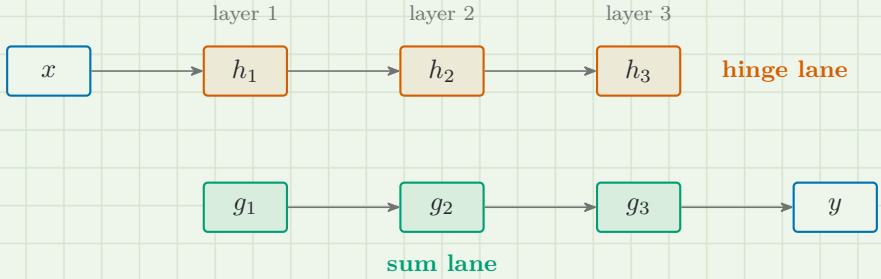
Two neurons per layer are enough, as the rest of the lecture shows. For functions of one variable, two is the least width that works (Hanin and Sellke, *Approximating continuous functions by ReLU nets of minimal width*, 2017).

Give each layer two ReLU neurons: on top a *hinge neuron* h_ℓ, which makes one kink, and below a *running-sum neuron* g_ℓ, which adds up the hinges handed down to it. Layer ℓ makes kink ℓ.



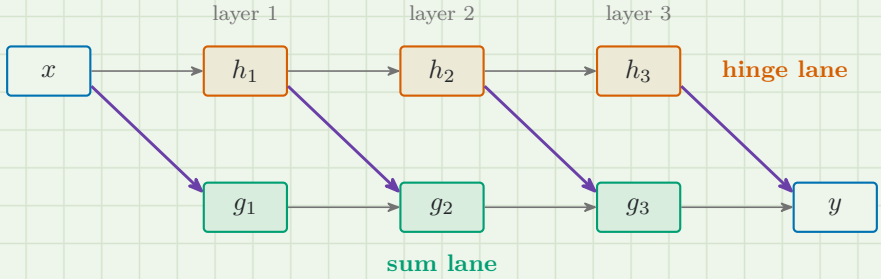
Hinge lane: each hinge neuron reads the hinge before it and makes the next kink.

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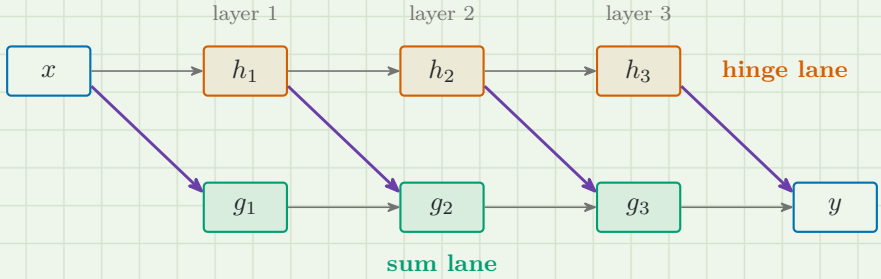
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Sum lane: each running sum passes its total on; the output y reads the last one.

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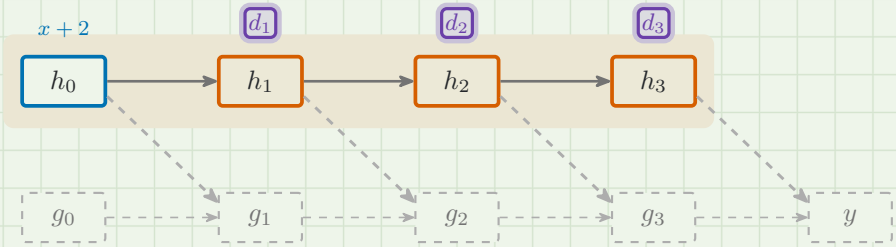


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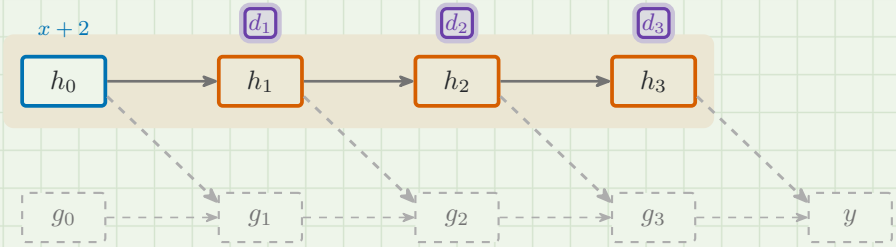
A chain of hinges above a chain of running sums: we call it a *hinge-chain network*.

Entry 5

The hinge lane



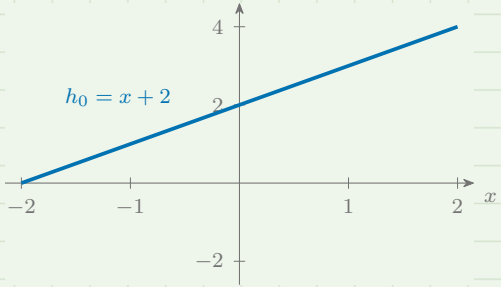
So far. A piecewise-linear function is its start value plus one hinge per kink. The plan: a hinge lane that makes the kinks, and a sum lane that adds them up.



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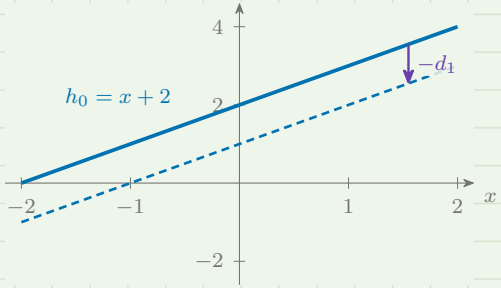
Now. The hinge lane. It starts from the ramp $h_0 = x + 2$, and each hinge neuron makes the next kink, one gap d_ℓ further to the right.

The hinge lane starts from the input shifted to 0 at the left end: the ramp $h_0 = x - a = x + 2$, which is hinge 0. The first hinge neuron lowers it by a *gap* $d_1 \geq 0$ and applies the floor: $h_1 = \mathrm{ReLU}(h_0 - d_1)$.



Blue: the ramp $h_0 = x + 2$, hinge 0, with its kink at the left end $a = -2$.

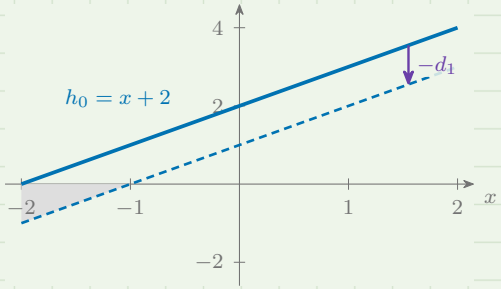
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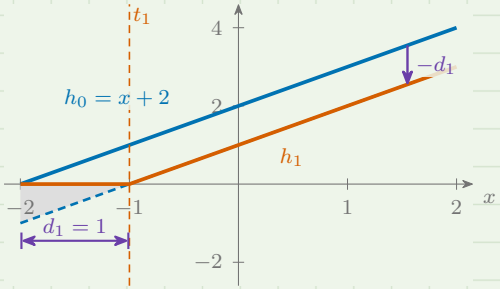


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The floor cuts off the part below zero (grey).

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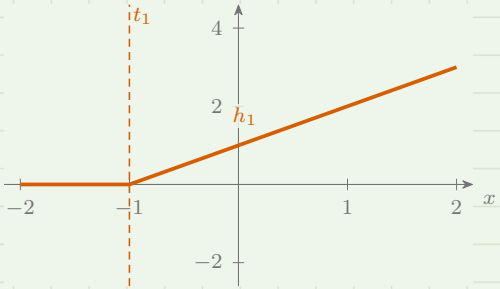
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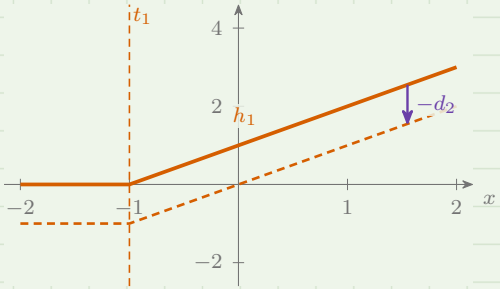
What is left is a hinge, $h_1 = \mathrm{ReLU}(x + 1)$, with its kink one gap to the right of a: at $t_1 = -1$.

The second hinge neuron lowers h_1 by the gap $d_2 = 1$ and applies the floor. The flat part sinks below zero and is cut back to 0: the hinge keeps its shape, and its kink moves right by d_2.



Start from the hinge h_1, kink at $t_1 = -1$.

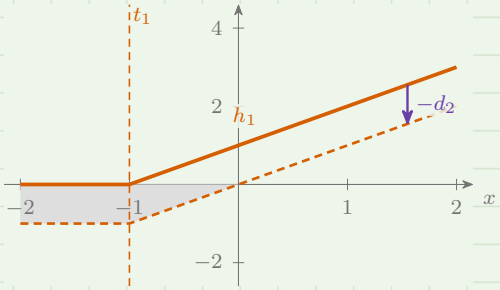
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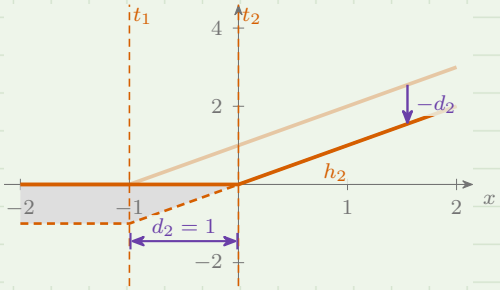


Start from the hinge h_1, kink at $t_1 = -1$.

Lower it by $d_2 = 1$: the flat part sinks to -1 (dashed).

The floor cuts off all that is below zero.

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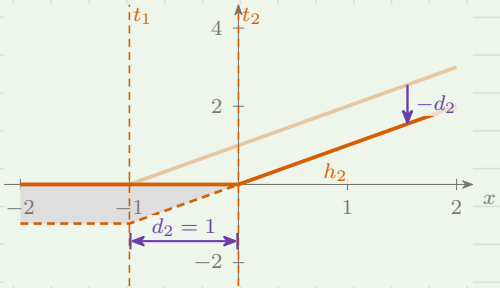
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The floor cuts off all that is below zero.

Left: $h_2 = \mathrm{ReLU}(x)$, the same hinge, its kink moved right by d_2 to $t_2 = 0$.

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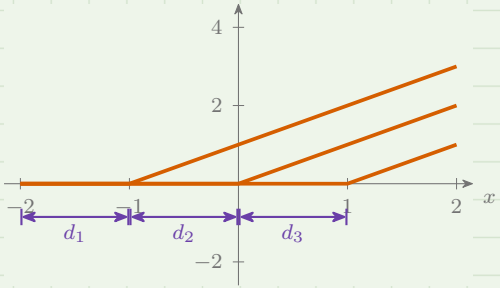
The floor cuts off all that is below zero.

Left: $h_2 = \mathrm{ReLU}(x)$, the same hinge, its kink moved right by d_2 to $t_2 = 0$.

$$h_\ell = \mathrm{ReLU}(h_{\ell-1} - d_\ell)$$

Every hinge neuron: kink $t_\ell = t_{\ell-1} + d_\ell$.

The kinks form a chain of steps from the left end a: kink ℓ sits ℓ gaps to the right of a. Make one gap longer, and every kink after it moves along.

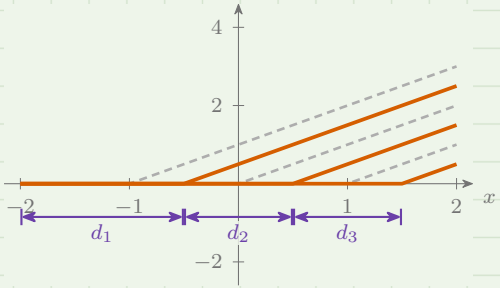


Vermillion: the hinges h_1, h_2, h_3. Violet: the gaps.

$$t_\ell = a + d_1 + d_2 + \cdots + d_\ell$$

The zigzag: $d = (1, 1, 1)$, so the kinks sit at -1, 0 and 1.

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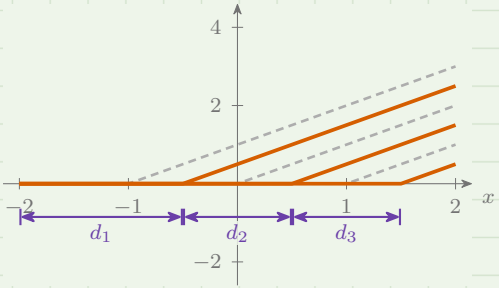
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Make the first gap longer, $d_1 = 1.5$: all three kinks move right by 0.5. The grey hinges are where they were.

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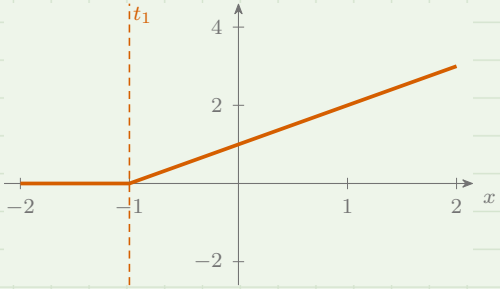
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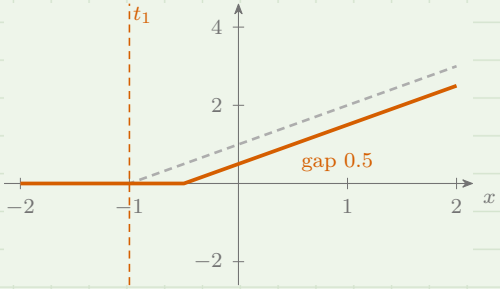
A gap is never negative, so the kinks stay in order. On the page, dragging kink ℓ in the plot sets its gap d_ℓ.

A gap lowers the hinge before it, and the floor cuts the sunken flat part back to 0. A negative gap would raise the hinge instead: its flat part stays above zero, the floor cuts nothing, and what comes out is no hinge at all.



Start from the hinge h_1, kink at -1.

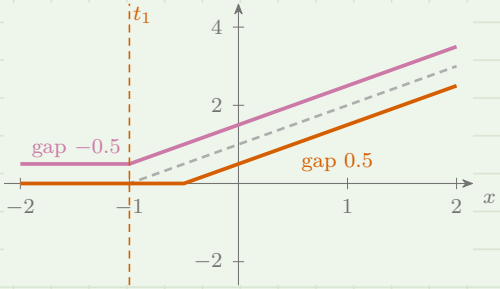
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Start from the hinge h_1, kink at -1.

A gap of 0.5: a hinge again, its kink moved right to -0.5.

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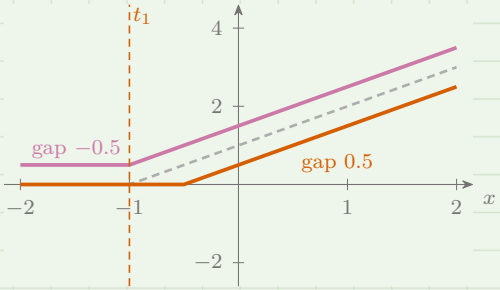


Start from the hinge h_1, kink at -1.

A gap of 0.5: a hinge again, its kink moved right to -0.5.

A gap of -0.5 raises the hinge by 0.5. The floor cuts nothing, and the flat part stays at 0.5. A hinge is 0 left of its kink, so this is no hinge.

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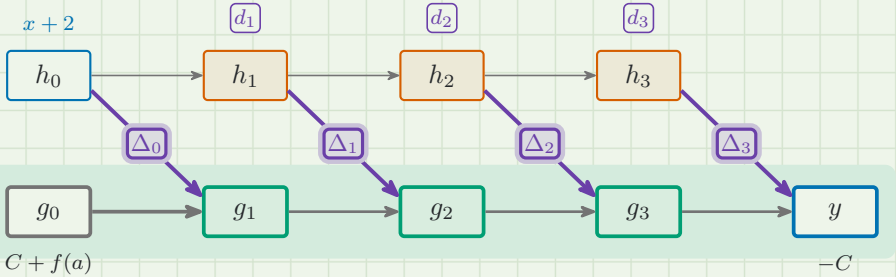
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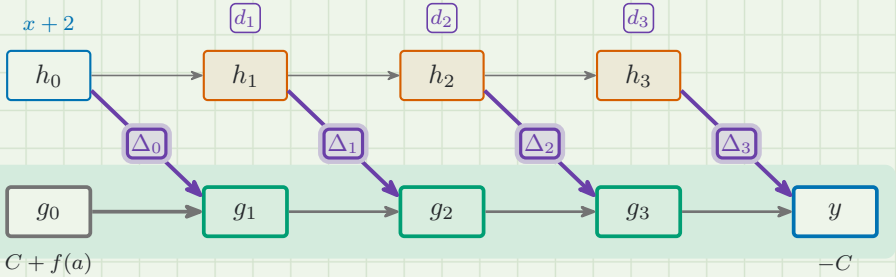
So every gap is at least 0, and the kinks come in order. The page keeps each gap between 0 and 4.

Entry 6

The sum lane



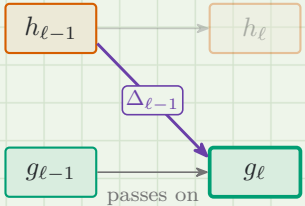
So far. The hinge lane makes one hinge per layer, $h_\ell = \mathrm{ReLU}(x - t_\ell)$, each kink one gap d_ℓ after the one before.



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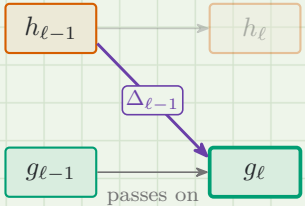
Now. The sum lane. Its running sums add the hinges up, each weighted by its slope change Δ_ℓ; a lift C keeps them above the floor; the output y adds the last hinge.

The sum lane keeps a running total of the hinge form, starting from the start value, $g_0 = f(a)$. Running sum g_ℓ passes the total $g_{\ell-1}$ on and adds the hinge before it, $h_{\ell-1}$, weighted by its slope change $\Delta_{\ell-1}$.



Grey arrow: the total passes on with weight 1.
Violet arrow: the hinge comes in, weighted by its slope change.

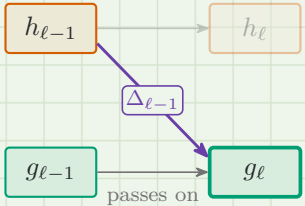
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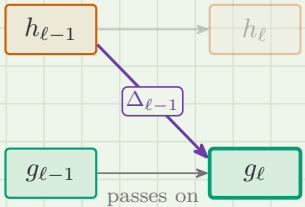


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For now, suppose that no total dips below zero, so the floor does nothing.

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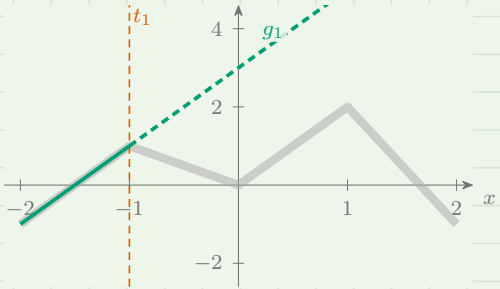
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For now, suppose that no total dips below zero, so the floor does nothing.

The first one, for the zigzag:
$g_1 = -1 + 2\,(x+2) = 2x + 3$, its first piece.

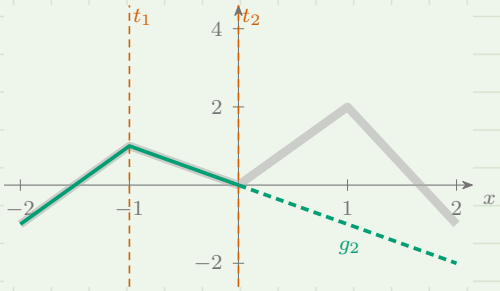
Running sum g_ℓ holds the start value and the hinges 0 to $\ell - 1$. So it follows the zigzag up to the kink t_ℓ, where the next hinge would bend it, and goes on straight from there.



Grey: the zigzag. Green: one running sum per click, dashed where it leaves the zigzag.

g_1: the start value and hinge 0. It is the zigzag up to $t_1 = -1$; then the line $2x + 3$ goes straight on.

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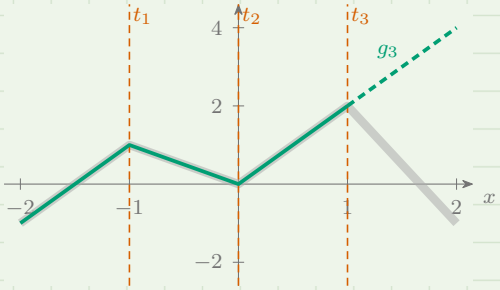


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g_1: the start value and hinge 0. It is the zigzag up to $t_1 = -1$; then the line $2x + 3$ goes straight on.

g_2 adds hinge 1: the zigzag up to $t_2 = 0$, then straight on with the slope -1.

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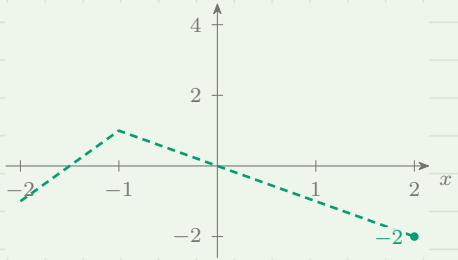
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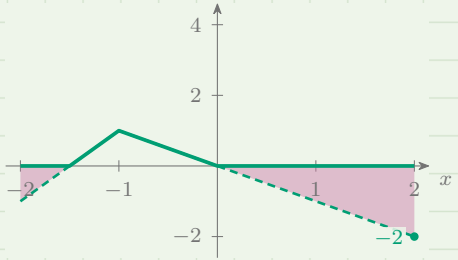
g_3 adds hinge 2: the zigzag up to $t_3 = 1$, then straight on with the slope 2. The last hinge comes in at the output.

The running sums are ReLU neurons too, so each has a floor at zero. The zigzag's totals dip below it: g_2 should hold -1 at the left end and -2 at the right end. The floor cuts those parts off, and nothing later brings them back.



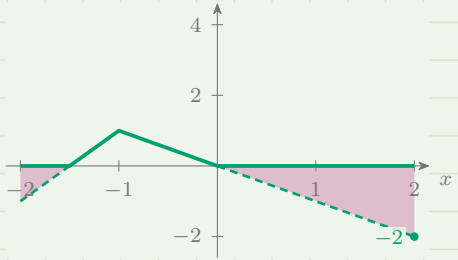
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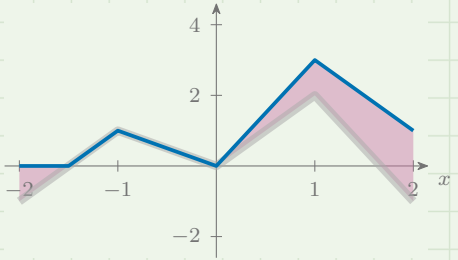


Dashed: the total g_2 should hold. Solid: what passes the floor. Pink: cut off and lost.

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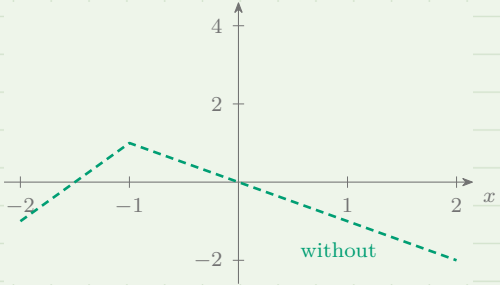


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With no lift, the output y misses the zigzag (grey) wherever a total was cut: pink, the error.

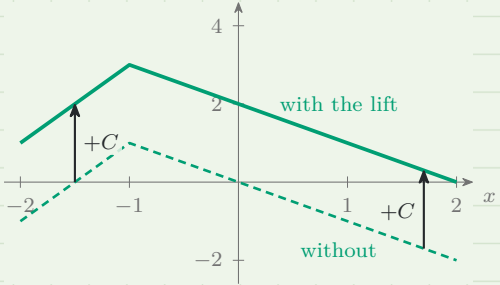
Start the sum lane with a deposit, a *lift* $C \geq 0$. Every total is then C higher and stays above the floor; the output pays C back. The deepest dip is -2, so the zigzag needs $C = 2$.



Green: the running sum g_2, dashed without the lift, solid with $C = 2$.

Without a lift, g_2 would dip to -2 at $x = 2$.

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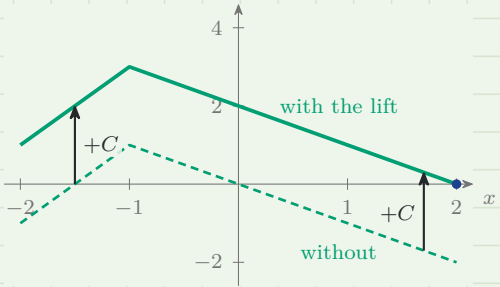
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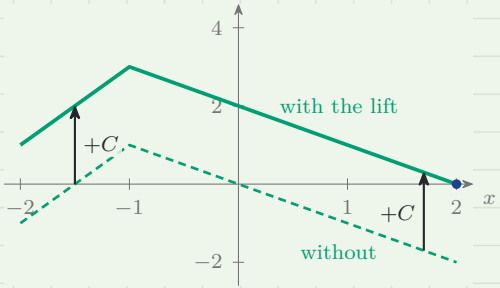
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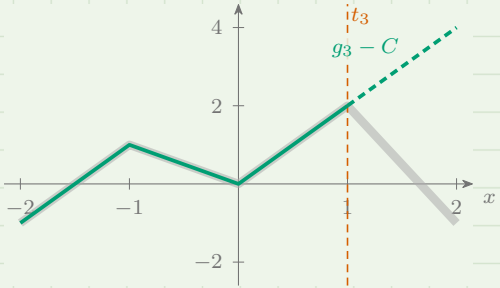
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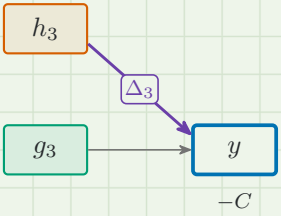
With $C = 2$, g_2 stays at or above zero, and so do g_1 and g_3.

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Moved right by 2 and up by 3, the same zigzag lives on $[0, 4]$ above the floor: no total dips there, and $C = 0$.

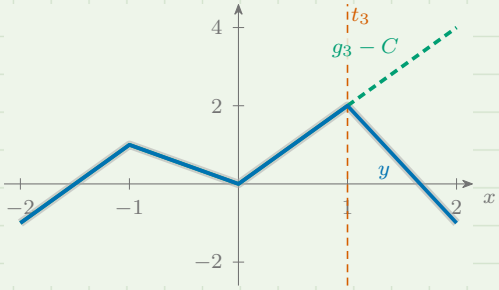
After the last layer, g_3 holds the zigzag up to the last kink, lifted by C. The output adds the last hinge, weighted by $\Delta_3 = -5$, and takes C off. It has no floor: y is a plain sum.



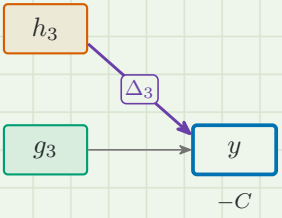


Green: $g_3 - C$.

After the last layer, g_3 holds the zigzag up to the last kink, lifted by C. The output adds the last hinge, weighted by $\Delta_3 = -5$, and takes C off. It has no floor: y is a plain sum.



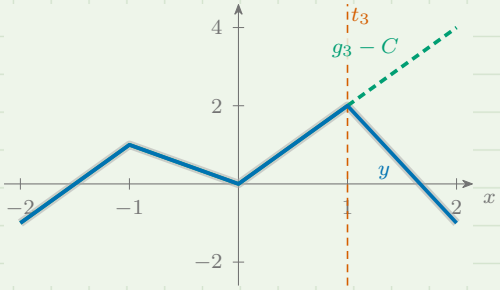
Green: $g_3 - C$. Blue: the output y, on top of the zigzag.



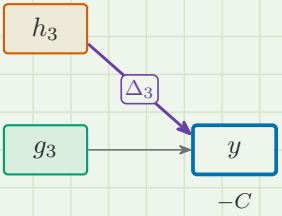
$$y = \Delta_k\, h_k + g_k - C$$

At $t_3 = 1$ the slope turns from 2 to $2 + (-5) = -3$.

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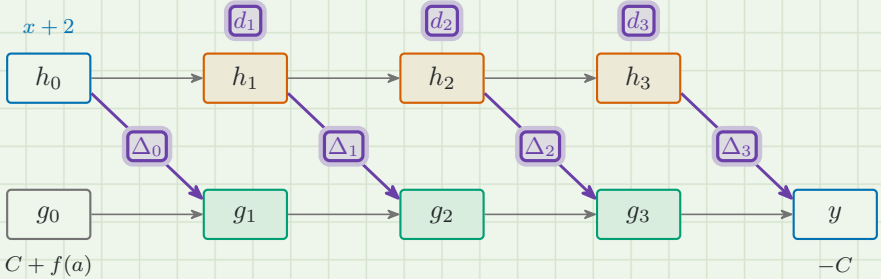
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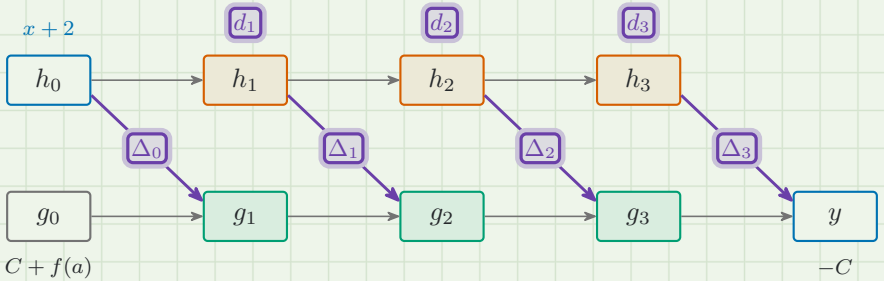
So $y = f$: the network draws the zigzag exactly.

Entry 7

The knobs



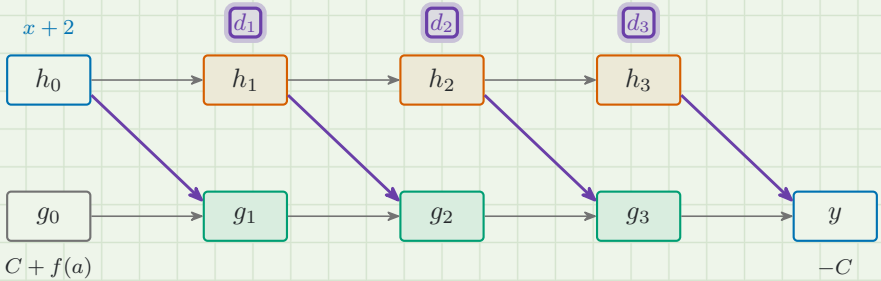
So far. The network draws the zigzag exactly: the hinge lane makes the kinks, the sum lane adds the hinges up from a lift C, and the output pays the lift back.



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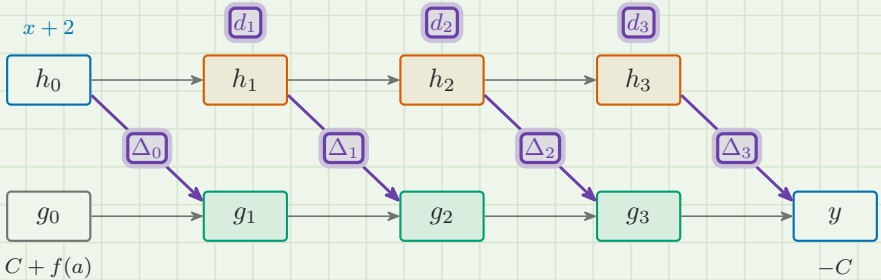
Now. Which numbers set the network? Only the gaps d_ℓ and the slope changes Δ_ℓ: the knobs.

Two kinds of numbers set the whole network: the intercept of each hinge, its gap d_ℓ, and the slope change after each hinge's ReLU, Δ_ℓ, the weight on its arrow down to the sum lane. They are the *knobs*.



Gaps: one in each hinge neuron, $d_1, \ldots, d_k$. Each places a kink, one gap after the kink before.

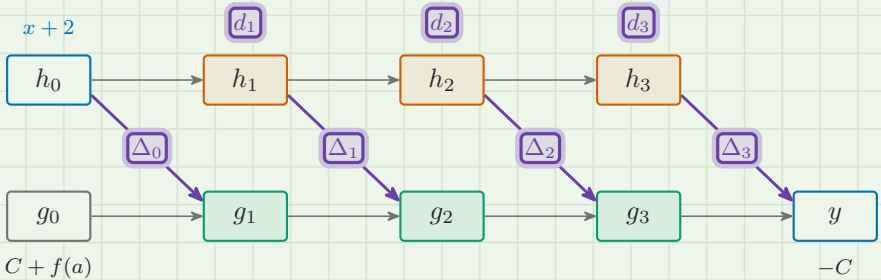
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Gaps: one in each hinge neuron, $d_1, \ldots, d_k$. Each places a kink, one gap after the kink before.

Slope changes: one on each violet arrow, $\Delta_0, \ldots, \Delta_k$. Each bends the curve at its kink.

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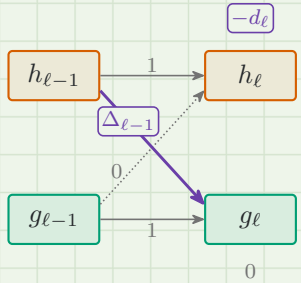


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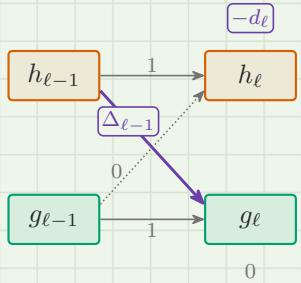
Slope changes: one on each violet arrow, $\Delta_0, \ldots, \Delta_k$. Each bends the curve at its kink.

In all, k gaps and $k+1$ slope changes: $2k+1$ knobs, seven for the zigzag's three layers.

Written the usual way, with a weight matrix and a bias, layer ℓ has two free entries. A hinge neuron reads the hinge before it with weight 1 and the sum with weight 0; a running sum passes its total on with weight 1.



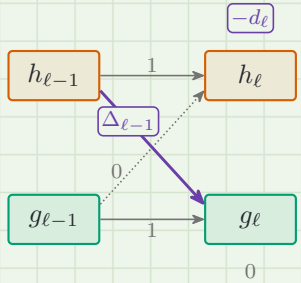
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$$\begin{pmatrix} h_\ell \\ g_\ell \end{pmatrix} = \mathrm{ReLU}\left(\begin{pmatrix} 1 & 0 \\ \Delta_{\ell-1} & 1 \end{pmatrix} \begin{pmatrix} h_{\ell-1} \\ g_{\ell-1} \end{pmatrix} + \begin{pmatrix} -d_\ell \\ 0 \end{pmatrix} \right)$$

Violet: the two knobs of the layer. Grey: fixed, the same for every target and every layer.

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Violet: the two knobs of the layer. Grey: fixed, the same for every target and every layer.

Two more numbers are no knobs: the start value $f(a)$, read off the target, and the lift C, the least that keeps every running sum above the floor. The page computes both.

Set the gaps to $d = (1, 1, 1)$ and the slope changes to $\Delta = (2, -3, 3, -5)$; then $f(a) = -1$ and $C = 2$ follow. Each box draws what its neuron outputs for x from -2 to 2.

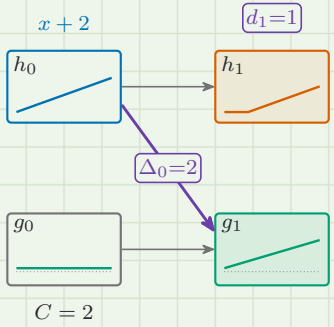
$x + 2$



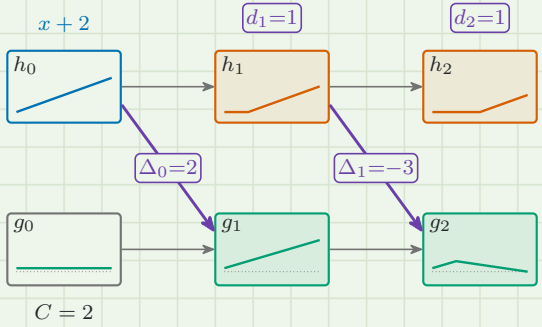


$C = 2$

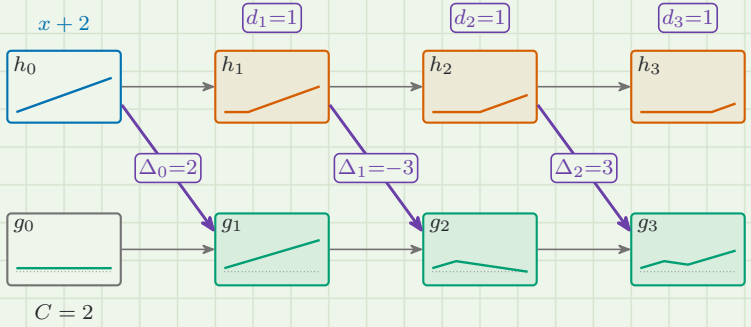
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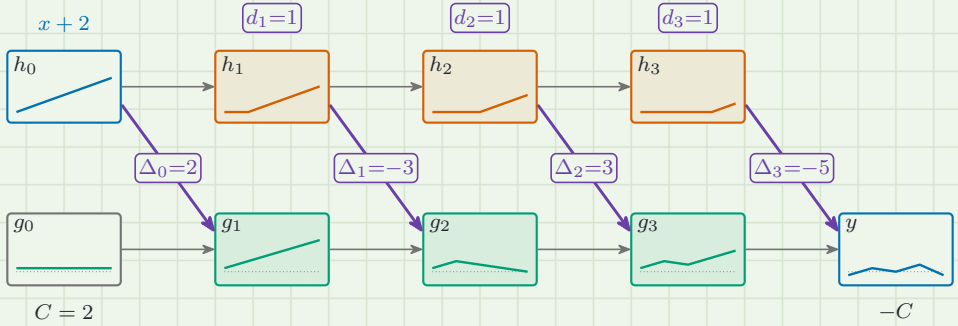
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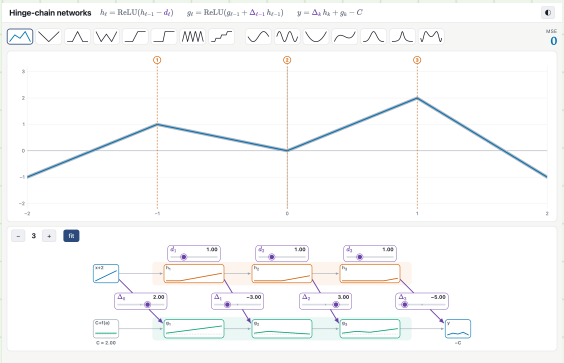


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Top: the hinges, with their kinks at -1, 0 and 1. Bottom: the running sums, each the zigzag so far, lifted by 2. Right: the output y, the zigzag.

The page *Hinge-chain networks* draws this network for fifteen targets: the plot on top, the network below, and every knob a violet slider where its number acts.

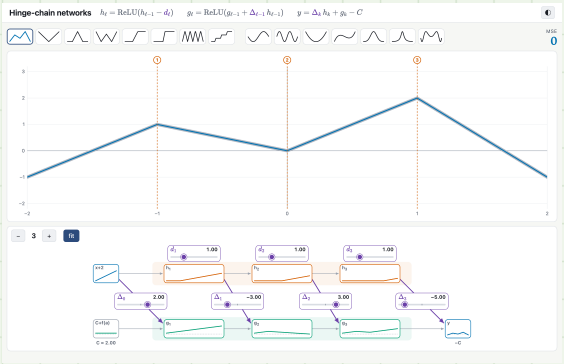


The zigzag in three layers, after fit: the error is 0.

Pick a target from the menu. It starts flat, at $f(a)$.

Open it in a browser: `eth-aml-hs26.github.io/neural-networks/viz/hinge-chain-networks`

The page *Hinge-chain networks* draws this network for fifteen targets: the plot on top, the network below, and every knob a violet slider where its number acts.



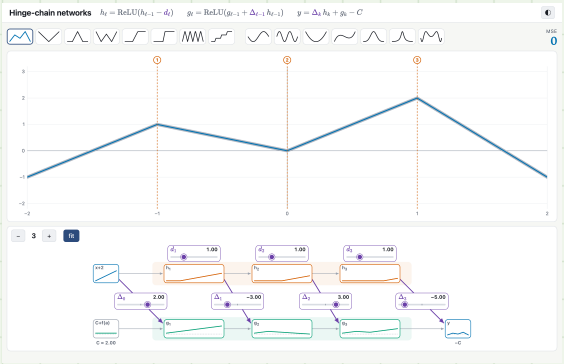
The zigzag in three layers, after fit: the error is 0.

Pick a target from the menu. It starts flat, at $f(a)$.

The buttons + and − add and remove a layer.

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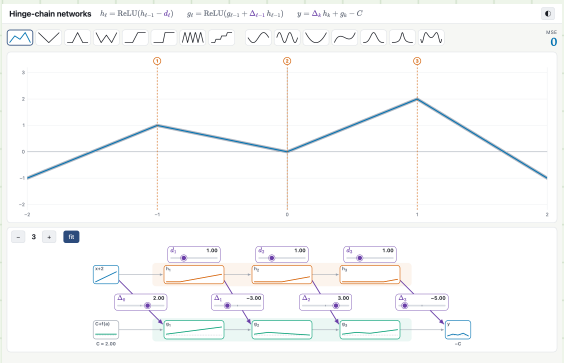
Pick a target from the menu. It starts flat, at $f(a)$.

The buttons + and − add and remove a layer.

Move a slider, or drag a kink in the plot to set its gap.

Open it in a browser: `eth-aml-hs26.github.io/neural-networks/viz/hinge-chain-networks`

The page *Hinge-chain networks* draws this network for fifteen targets: the plot on top, the network below, and every knob a violet slider where its number acts.



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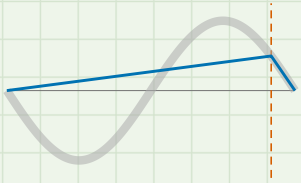
The buttons + and − add and remove a layer.

Move a slider, or drag a kink in the plot to set its gap.

fit sets every knob by the construction. A piecewise-linear target with k kinks is matched exactly by k layers.

Open it in a browser: `eth-aml-hs26.github.io/neural-networks/viz/hinge-chain-networks`

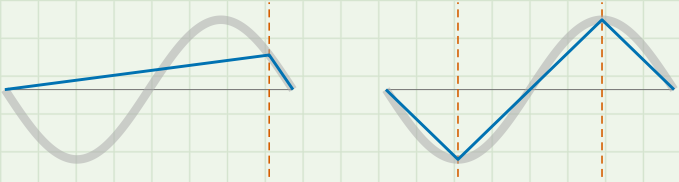
A sine has no kinks to copy. With k layers, *fit* picks the k kinks whose straight pieces follow the sine most closely. The mean squared error falls from 0.42 with one layer to 2.3×10^{-4} with ten.



$k = 1$: error 0.42

Grey: the sine. Blue: the network after fit. Dashed: its kinks.

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$k = 1$: error 0.42 $k = 2$: error 0.023

Grey: the sine. Blue: the network after fit. Dashed: its kinks.

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$k = 1$: error 0.42 $k = 2$: error 0.023 $k = 4$: error 0.0040

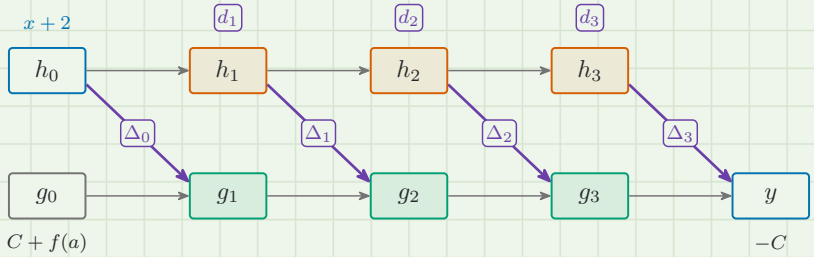
Grey: the sine. Blue: the network after fit. Dashed: its kinks.

Each layer adds one kink, so the network bends where the sine bends most. On the page, press + and then fit, and watch the error fall.

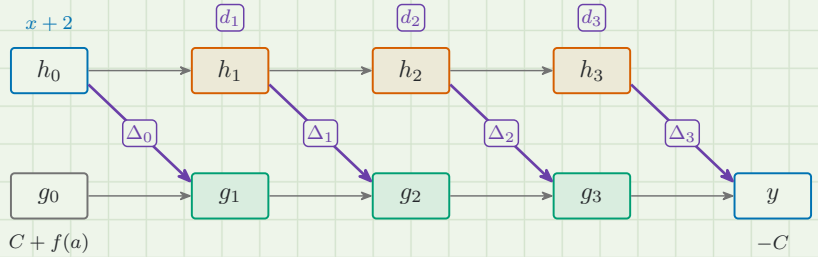
Entry 8

Summary

A hinge-chain network draws any continuous piecewise-linear function exactly. Its top lane makes the hinges, one kink per layer; its bottom lane adds them up.

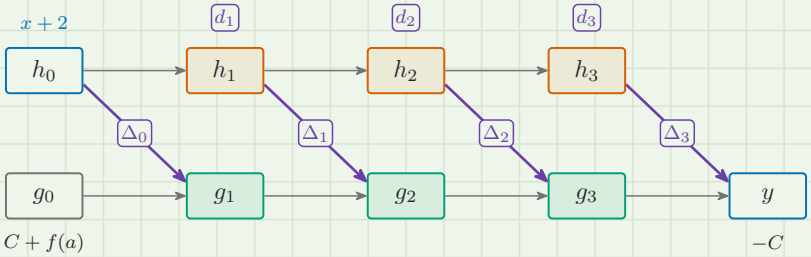


A hinge-chain network draws any continuous piecewise-linear function exactly. Its top lane makes the hinges, one kink per layer; its bottom lane adds them up.



$$h_\ell = \mathrm{ReLU}(h_{\ell-1} - d_\ell) \qquad g_\ell = \mathrm{ReLU}(g_{\ell-1} + \Delta_{\ell-1}\, h_{\ell-1}) \qquad y = \Delta_k\, h_k + g_k - C$$

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The knobs are the gaps d_ℓ and the slope changes Δ_ℓ, $2k+1$ numbers. Every other weight is 1 or 0, and $f(a)$ and the lift C follow from the target.